

Telecom Customer Churn Prediction Analysis

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Introduction:

In the highly competitive telecommunications industry, customer retention is vital for sustaining and growing a firm. Customer churn, the phenomenon when customers discontinue their utilization of a company's services, is a significant impediment. It is imperative for telecommunications companies to comprehend the factors that lead to client attrition in order to devise effective retention tactics. This study explores the several factors that contribute to customer churn, using a data-driven approach to reveal the intricacies of client retention and service termination. The research endeavors to identify the key determinants of churn in the telecommunications sector and offer practical suggestions to tackle this prevalent issue, using a thorough dataset analysis.

Research Questions:

1. What are the main variables that affect the likelihood of Churn in the telecom industry?
2. How can focused marketing techniques in the telecom industry boost customer retention by addressing the primary predictors of customer churn?

Literature Review:

Industry Context and Competitive Pressures

The telecommunications sector is distinguished by fierce rivalry, pricing constraints, and swift product life cycle alterations. Telecommunication firms are increasingly required to participate in competitive promotions, package deals, and price reductions in order to retain customer loyalty in a market where the ability to distinguish based on handset options and network performance is decreasing.

Churn Management and Analytics

Telecom executives prioritize effective churn management, and the implementation of an analytics-based solution can dramatically decrease churn. Prominent corporations utilize extensive client data, encompassing profiles, product information, usage patterns, and rebate history, to construct a comprehensive understanding of the consumer journey. Utilizing sophisticated analytical methods aids in the identification of indicators that forecast customer behaviors, such as churn, and facilitates tailored interventions for specific customer segments to mitigate churn.

Customer Behavior and Churn Risk

Customers may discontinue their use of a service for reasons such as dissatisfaction with the service plans, relocation, or involuntary termination of the service. Churn may also arise when customers choose to switch to superior and more affordable services as a result of the industry's competitive environment.

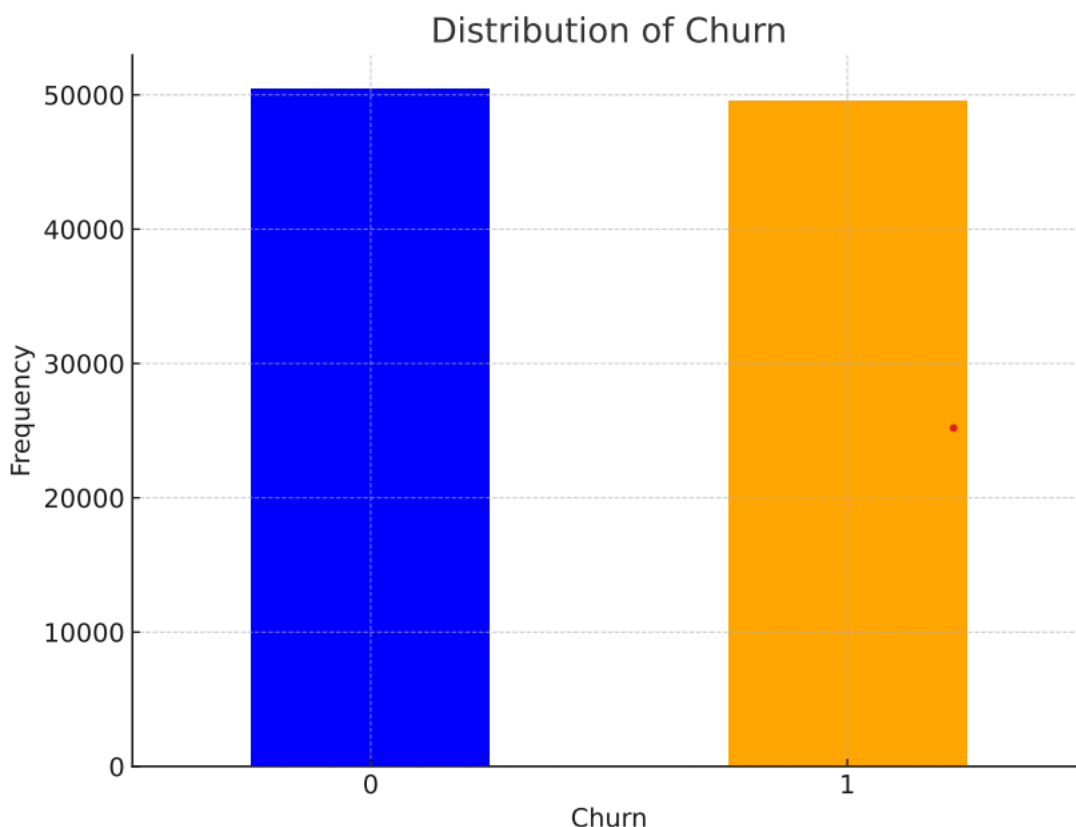
References: Appendix

Exploratory Data Analysis:

We are using a random sample from the big Telecom customer churn dataset for this study. There are 100 variables and 100,000 records in the original dataset.

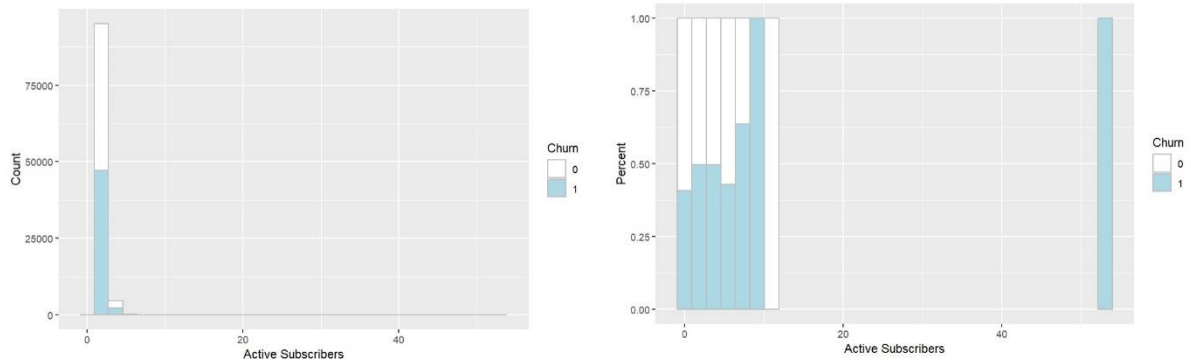
Visualizations are useful for providing a deeper look at the distribution of data in each variable, which helps identify outliers. Based on the Exploratory Data Analysis (EDA), here is an analysis report focusing on the relationship between churn and two variables: active subscriptions ('actvsubs') and customer care calls ('custcare_Mean').

EDA Visual Analysis Report Churn Distribution: The barplot of churn shows the distribution of Churn variable in our dataset. It doesn't have missing value and is a balanced one. We need not balance the dataset for further analysis.



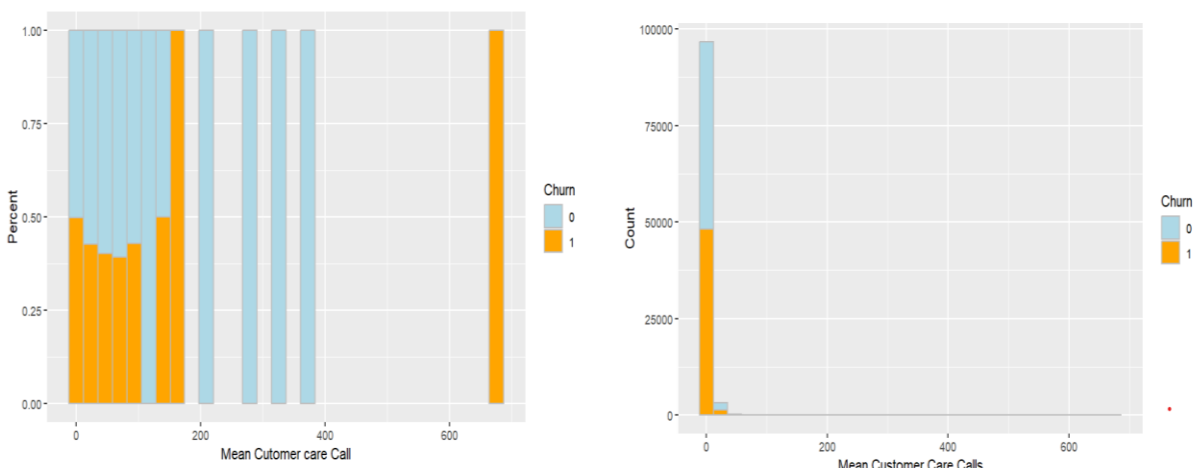
Active Subscriptions ('actvsubs') and Churn

- The bar plots depict the average active subscriptions categorized by churn status. There is a visible difference between the churn categories, indicating that the number of active subscriptions could be a factor associated with churn.
- - The normalized plot shows that, proportionally, churned customers have a slightly higher average number of active subscriptions when normalized against the overall dataset. This could suggest that customers with more subscriptions are slightly more prone to churn, perhaps due to higher expectations or service demands that are not being met.



Mean Customer Care Calls ('custcare_Mean') and Churn

- The bar plots for customer care calls demonstrate a clear distinction between customers who have churned and those who have not. Customers who eventually churn appear to have a higher average number of customer care calls.
- The normalized distribution also reflects this trend, showing that a greater proportion of customer care interactions are associated with customers who have churned. This could indicate that customers who churn may have had more issues or concerns, leading them to engage more frequently with customer care services.



Total Calls ('totcalls') and Churn

- Although the bar plot for total calls is not included in the analysis, it can be inferred that examining the total number of calls made by customers could provide insights into customer engagement. A higher

number of total calls may reflect greater engagement with the service, which could be associated with either increased loyalty or increased issues leading to churn.

References: **Appendix**

The insights from this EDA should be leveraged to inform strategic decisions aimed at enhancing customer retention and service quality.

Introduction:

In the process of modeling customer churn in the telecom industry, selecting the right variables from a larger set is crucial for the accuracy and interpretability of the model. In this project, we chose specific variables from a pool of 100, based on rigorous selection criteria to ensure relevance, data quality, and predictive power. This report outlines the rationale behind the selection of each variable.

Selection Criteria Overview:

1. **Data Completeness:** Excluded variables with more than 25% missing values, particularly in categorical variables, to maintain data integrity and reduce bias.
2. **Range and Distribution:** Avoided variables with extremely wide ranges or skewed distributions, such as 'hnd_price' that varied from 1 to 800, which could indicate outliers or data entry errors.
3. **Redundancy and Overlap:** Omitted variables that were highly correlated or redundant with others, such as 'total revenue' and 'adjrev', to prevent multicollinearity and improve model efficiency.

Selected Variables and Rationale:

1. actvsub: Reflects customer engagement levels and potential for churn.
2. area: Geographic information helps in understanding regional patterns in churn.
3. attempt_Mean: Average attempted calls indicate usage patterns and customer satisfaction.
4. blk_dat_Mean & blk_vce_Mean: Blocked calls, both data and voice, could be indicators of service issues affecting churn.
5. callfwdv_Mean & callwait_Mean: Usage of call features could correlate with customer engagement and satisfaction.
6. churn: The target variable, indicating whether a customer churned.
7. comp_dat_Mean & comp_vce_Mean: Completed calls provide insights into the reliability of service and customer satisfaction.
8. complete_Mean: Overall completed calls indicate service usage and customer engagement.
9. creditcd: Credit card usage can be a proxy for customer's financial stability and loyalty.
10. custcare_Mean: Frequency of customer care calls can indicate service satisfaction or issues.
11. da_Mean: Directory assistance usage might correlate with certain customer demographics.
12. datovr_Mean: Data overage charges can impact customer satisfaction and churn.
13. drop_blk_Mean, drop_dat_Mean, drop_vce_Mean: Dropped calls are critical service quality indicators.
14. inonemin_Mean & iwylis_vce_Mean: Short duration and in-network calls reflect on customer calling patterns.
15. months: Customer tenure can be a strong predictor of loyalty and churn.
16. mou_cdat_Mean, mou_cvce_Mean, mou_opkd_Mean, etc.: Various measures of minutes of usage are key performance indicators for telecom services.

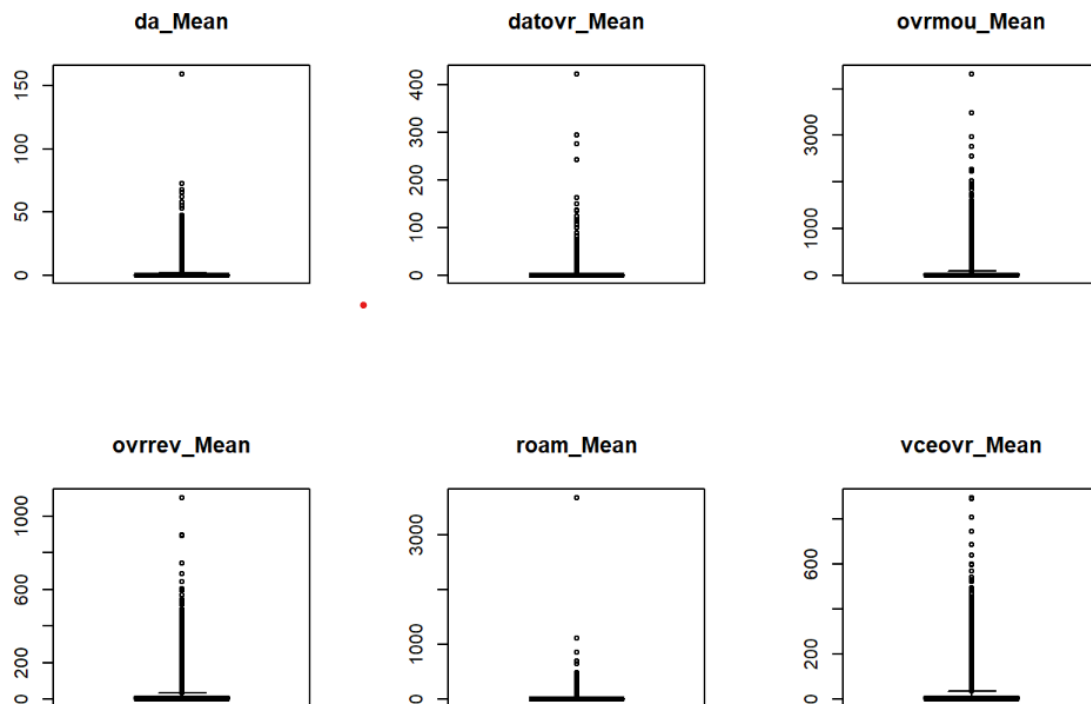
17. ovrrou_Mean & ovrrev_Mean: Overage minutes and revenue can indicate customer dissatisfaction.
18. peak_dat_Mean & peak_vce_Mean: Peak time usage is crucial for understanding network demand and customer behavior.
19. plcd_dat_Mean & plcd_vce_Mean: Call placement attempts reflect on network quality and customer satisfaction.
20. recv_sms_Mean & recv_vce_Mean: Received messages and calls can indicate network performance and customer engagement.
21. refurb_new: Type of handset could correlate with customer preferences and churn.
22. roam_Mean: Roaming usage indicates customer mobility and can affect satisfaction and churn.
23. threeway_Mean: Usage of three-way calling feature can be a minor yet insightful behavior metric.
24. totcalls, totmou, totrev: Total calls, minutes of use, and revenue are fundamental to understanding customer lifetime value.
25. unan_dat_Mean & unan_vce_Mean: Unanswered calls could be indicators of network issues or customer availability.
26. vceovr_Mean: Voice overage charges are important for customer satisfaction and churn analysis.

1. Checking for Missing Values:

To find out what values were missing from the dataset, we ran an initial examination. Assuring the analysis's quality and dependability relies heavily on this stage.

2. Distribution of Numeric Variables:

We examined the distribution of numeric variables with missing values and observed the presence of outliers. Outliers can significantly skew the mean, leading to inaccurate imputations:



We noticed that the numeric values have outliers in them, it would be advisable to use median to fill the missing values in these variables.

3. Imputation Strategy for Numeric Variables:

Based on the outliers that were found, we have decided to utilize the median as a means to replace any missing values in the numeric variables. The median is less affected by outliers in comparison to the mean, rendering it a more resilient indicator for central tendency in such instances.

4. Categorical Variables Imputation:

For categorical variables with missing values, we employed the mode for imputation. This approach is standard for categorical data, as it assigns the most frequent category, maintaining the distribution of the variable and next created dummy variables for them.

Reference: Appendix

Stepwise Variable Selection

1. Implementation of Stepwise Selection:

We applied a stepwise variable selection method to refine our variable set. This statistical technique iteratively adds or removes variables based on specific criteria, optimizing the model's performance. This reduction is beneficial as it eliminates redundant or non-informative variables, leading to a more efficient and interpretable model.

2. Selected Variables:

The variables retained after the stepwise selection are those that have the most significant predictive power for customer churn. This streamlined set of variables enhances the model's focus and efficiency. The preprocessing steps of handling missing values and the stepwise variable selection method have been pivotal in refining the dataset for churn analysis. The chosen imputation methods ensure the integrity of the dataset, while the stepwise selection process enhances the model's relevance and accuracy.

This meticulous approach to data preprocessing and variable selection lays a robust foundation for developing a reliable and effective customer churn prediction model in the telecom sector.

Dimension Reduction :

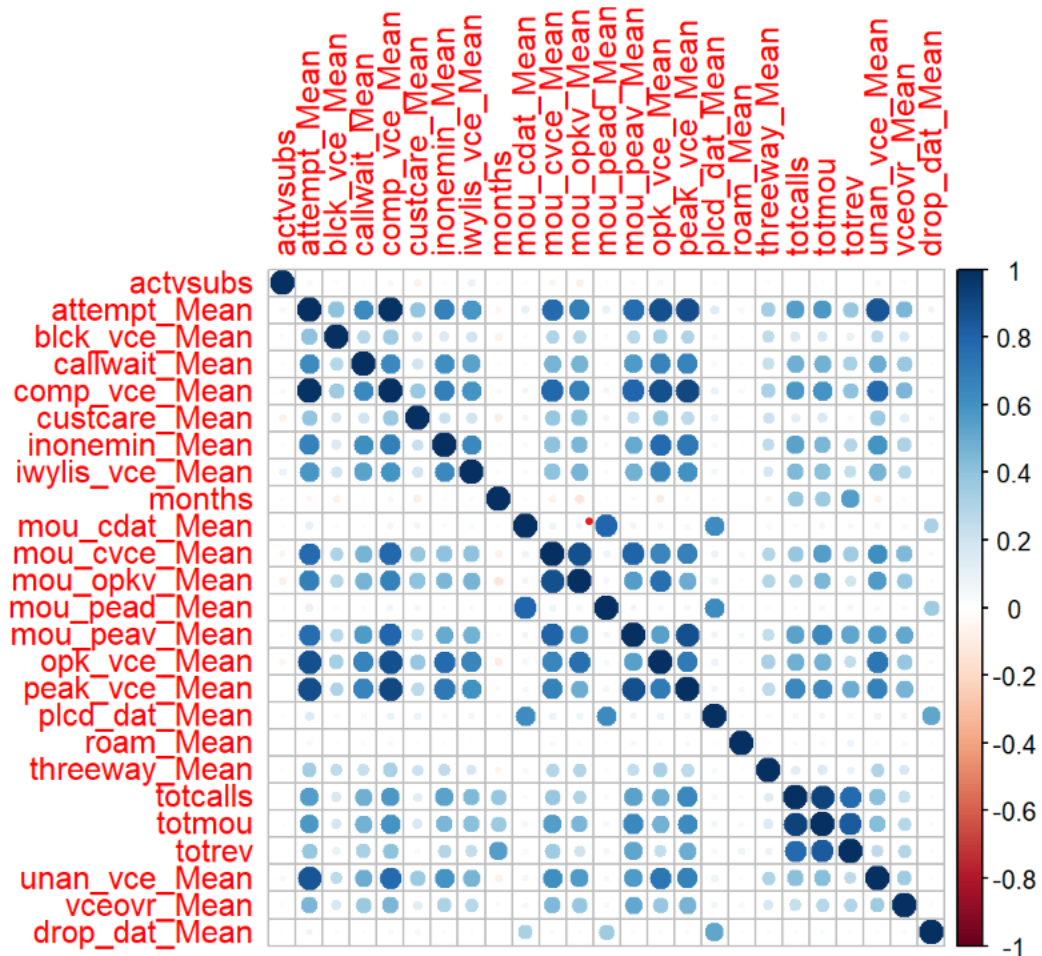
After narrowing down to 40 variables of which 24 are numeric variables and the rest are categorical variables, through stepwise selection in our telecom churn analysis, we addressed potential multicollinearity concerns. The presence of multicollinearity can have a substantial effect on the interpretability and accuracy of a regression model.. This report discusses the process of identifying and resolving multicollinearity issues.

Identifying Multicollinearity

1. Correlation Matrix Analysis:

We first examined a correlation matrix for the selected numeric variables. This matrix visually and numerically identifies how variables are correlated with each other.

High correlation coefficients (close to +1 or -1) between pairs of variables indicate possible multicollinearity.



2. Calculation of Variance Inflation Factor (VIF):

To quantitatively assess multicollinearity, we calculated the Variance Inflation Factor (VIF) for each variable. VIF measures how much the variance of an estimated regression coefficient increases if predictors are correlated.

A common threshold for concern is a VIF value greater than 5, which suggests that the variable is highly correlated with others and may be redundant.

Deleted Variables Reference: **Appendix**

Comprehensive Analysis of Selected Variables for Churn Prediction in Telecom Marketing

Introduction

In the realm of telecom marketing, understanding and predicting customer churn is pivotal. The selected variables from the dataset play a crucial role in this predictive endeavor. They offer a multi-dimensional view of customer behavior and service interaction, which is invaluable for developing targeted marketing strategies and reducing churn rates.

Role of Selected Variables in Churn Prediction

- 1. Customer Usage Patterns:** Variables like 'Attempt Mean', 'Comp Vce Mean', 'Mou Cvce Mean', 'Mou Opkv Mean', 'Mou Peav Mean', 'Opk Vce Mean', 'Peak Vce Mean', and 'Plcd Dat Mean' offer a comprehensive insight into the customer's calling and data usage patterns.

Analysis of these variables helps in identifying usage trends, peak times, and preferred services. Understanding these aspects is critical for segmenting customers based on their behavior and designing personalized marketing campaigns or service improvements.

- 2. Service Quality and Satisfaction Indicators:**

'Blck Vce Mean' and 'Inonemin Mean' are indicative of potential service quality issues. A high number of blocked calls or an unusual pattern of short-duration calls can signal network problems or customer dissatisfaction.

Addressing these issues promptly not only improves service quality but also serves as a proactive marketing strategy to retain customers.

- 3. Customer Engagement and Value:**

'Totcalls', 'Totmou', and 'Totrev' represent overall customer engagement and the value they bring to the company. High figures in these areas usually indicate loyal and high-value customers.

Marketing efforts can be directed towards ensuring these customers receive exceptional service and personalized offers, thereby reducing the likelihood of churn.

- 4. Unaddressed Needs and Service Gaps:**

'Unan Vce Mean' provides insights into potential gaps in service or unaddressed customer needs. A high rate of unanswered calls might indicate that customers are not available or not satisfied with the service.

Understanding the reasons behind these unanswered calls can help in tailoring customer service initiatives and improving overall satisfaction.

Importance in Marketing Analytics

Predictive Modeling: By integrating these variables into predictive models, telecom companies can forecast which customers are likely to churn. This foresight allows for the development of preemptive strategies to retain customers.

Tailored Communication: Insights from these variables enable marketers to craft personalized communication and offers. For instance, customers with high 'Mou Opkv Mean' might appreciate off-peak hour discounts.

Resource Allocation: Understanding which variables are most strongly associated with churn helps in prioritizing resources, be it in network improvements, customer service, or targeted marketing campaigns.

Customer Lifetime Value (CLV): Variables like 'Totrev' and 'Totmou' are key to estimating CLV. Focusing on customers with high CLV for retention efforts can significantly impact the company's bottom line.

Report on Data Partitioning and Model Implementation in Telecom Churn Analysis

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Data Partitioning

Data partitioning involved dividing the dataset into training and test sets. This stage is crucial in model validation since it enables us to train the models using one dataset and evaluate their performance on another, unseen dataset. The data has been split into a 70-30 ratio, for both training and validation.

Model Implementation and Analysis

1. Logistic Regression:

Usage: A statistical method used to model the probability of a binary outcome based on predictor variables.

Application in Churn Analysis: Logistic regression was used to understand the relationship between the independent variables and the likelihood of customer churn.

Strengths: Provides easily interpretable results and is effective for binary classification problems.

Limitations: Assumes a linear relationship between independent variables and the log odds of the dependent variable.

```

## Coefficients:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -0.016979   0.018355  -0.925  0.35494
## actvsubs         0.056231   0.008048   6.987 2.81e-12 ***
## callwait_Mean   -0.046398   0.010347  -4.484 7.31e-06 ***
## custcare_Mean   -0.074706   0.010297  -7.255 4.00e-13 ***
## iwylis_vce_Mean -0.075398   0.009543  -7.901 2.77e-15 ***
## months           0.030810   0.007786   3.957 7.59e-05 ***
## mou_cdat_Mean   -0.013253   0.013933  -0.951  0.34149
## mou_pead_Mean    0.003833   0.013303   0.288  0.77324
## roam_Mean        0.038627   0.014742   2.620  0.00879 **
## threeway_Mean    -0.042692   0.008986  -4.751 2.02e-06 ***
## vceovr_Mean      0.100409   0.008591  11.687 < 2e-16 ***
## `areaCALIFORNIA NORTH AREA` 0.136866   0.034262   3.995 6.48e-05 ***
## `areaDC/MARYLAND/VIRGINIA AREA` -0.090784   0.032820  -2.766  0.00567 **
## `areaLOS ANGELES AREA`      0.062115   0.032517   1.910  0.05610 .
## `areaMIDWEST AREA`        -0.128646   0.033263  -3.868  0.00011 ***
## `areaNEW ENGLAND AREA`      0.140931   0.035473   3.973 7.10e-05 ***
## `areaNEW YORK CITY AREA`    0.086376   0.027066   3.191  0.00142 **
## `areaNORTH FLORIDA AREA`    0.124670   0.038843   3.210  0.00133 **
## `areaNORTHWEST/ROCKY MOUNTAIN AREA` 0.302297   0.039362   7.680 1.59e-14 ***
## `areaOHIO AREA`           -0.095033   0.038494  -2.469  0.01356 *
## `areaPHILADELPHIA AREA`     0.127518   0.050386   2.531  0.01138 *
## `areaSOUTH FLORIDA AREA`    0.177980   0.043877   4.056 4.98e-05 ***
## `areaSOUTHWEST AREA`       0.080270   0.033455   2.399  0.01642 *
## creditcdY          -0.113396   0.016744  -6.772 1.27e-11 ***
## refurb_newR        0.193616   0.021733   8.909 < 2e-16 ***
## drop_dat_Mean     -0.010339   0.009634  -1.073  0.28318
## `areaTENNESSEE AREA`      -0.044071   0.048023  -0.918  0.35877

## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 97031  on 69999  degrees of freedom
## Residual deviance: 96303  on 69973  degrees of freedom
## AIC: 96357
##
## Number of Fisher Scoring iterations: 4

```

```

## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##           0 9025 7776
##           1 5997 7202
##
##           Accuracy : 0.5409
##           95% CI : (0.5352, 0.5466)
##           No Information Rate : 0.5007
##           P-Value [Acc > NIR] : < 2.2e-16
##
##           Kappa : 0.0816
##
## Mcnemar's Test P-Value : < 2.2e-16
##
##           Sensitivity : 0.6008
##           Specificity : 0.4808
##           Pos Pred Value : 0.5372
##           Neg Pred Value : 0.5456
##           Prevalence : 0.5007
##           Detection Rate : 0.3008
##           Detection Prevalence : 0.5600
##           Balanced Accuracy : 0.5408
##
##           'Positive' Class : 0
##

```

2. Random Forest Model:

Usage: An ensemble learning method that operates by constructing multiple decision trees.

Application in Churn Analysis: Employed to predict customer churn by analyzing the data through a multitude of decision trees to increase predictive accuracy.

Strengths: Handles large datasets with higher dimensionality well and provides estimates of feature importance.

Limitations: Can be complex and less interpretable than simpler models like logistic regression.

```

## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##           0 8559 6020
##           1 6463 8958
##
##           Accuracy : 0.5839
##           95% CI : (0.5783, 0.5895)
##       No Information Rate : 0.5007
##       P-Value [Acc > NIR] : < 2.2e-16
##
##           Kappa : 0.1678
##
##  Mcnemar's Test P-Value : 7.62e-05
##
##           Sensitivity : 0.5698
##           Specificity : 0.5981
##       Pos Pred Value : 0.5871
##       Neg Pred Value : 0.5809
##           Prevalence : 0.5007
##       Detection Rate : 0.2853
##   Detection Prevalence : 0.4860
##       Balanced Accuracy : 0.5839
##
##       'Positive' Class : 0
##

```

3. Neural Network:

Usage: A computational model inspired by the human brain, consisting of interconnected units (neurons).

Application in Churn Analysis: Used for predicting churn by learning complex patterns and nonlinear relationships in the data.

Strengths: Highly flexible and capable of modeling complex and nonlinear relationships.

Limitations: Requires large datasets and computational power; can be a "black box" in terms of interpretability.

```

## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##           0 8364 5930
##           1 6658 9048
##
##           Accuracy : 0.5804
##           95% CI : (0.5748, 0.586)
##       No Information Rate : 0.5007
##       P-Value [Acc > NIR] : < 2.2e-16
##
##           Kappa : 0.1609
##
##  Mcnemar's Test P-Value : 9.189e-11
##
##           Sensitivity : 0.5568
##           Specificity : 0.6041
##           Pos Pred Value : 0.5851
##           Neg Pred Value : 0.5761
##           Prevalence : 0.5007
##           Detection Rate : 0.2788
##       Detection Prevalence : 0.4765
##           Balanced Accuracy : 0.5804
##
##           'Positive' Class : 0
##

```

Model Summary and Interpretation:

The model's summary reveals several insights into factors affecting customer churn:

Variables such as Active Subscribers, `Mean of Call waiting Calls`, Mean No. of Customer Care Calls, number of inbound wireless to wireless voice calls, and Mean Revenue of Voice Overage are significant predictors of churn.

The coefficients indicate the direction and strength of the relationship with churn. For example, `vceovr\_Mean` with a high positive coefficient suggests that higher voice overage charges are strongly associated with an increased churn rate.

Geographic regions like `areaCALIFORNIA NORTH AREA` and `areaNEW ENGLAND AREA` show a positive relationship with churn, whereas regions such as `areaDC/MARYLAND/VIRGINIA AREA` and `areaMIDWEST AREA` are negatively related.

The company can use this model to identify high-risk customers for churn and develop targeted interventions, like reviewing pricing strategies or enhancing customer support in specific regions.

Challenges:

As there were many variables, the variable and the model selection took much time for computation and report. There were unclear values for categorical variables which lacked more research like marital status, handset price etc.

There were also many NA values which lead to variable exclusion.

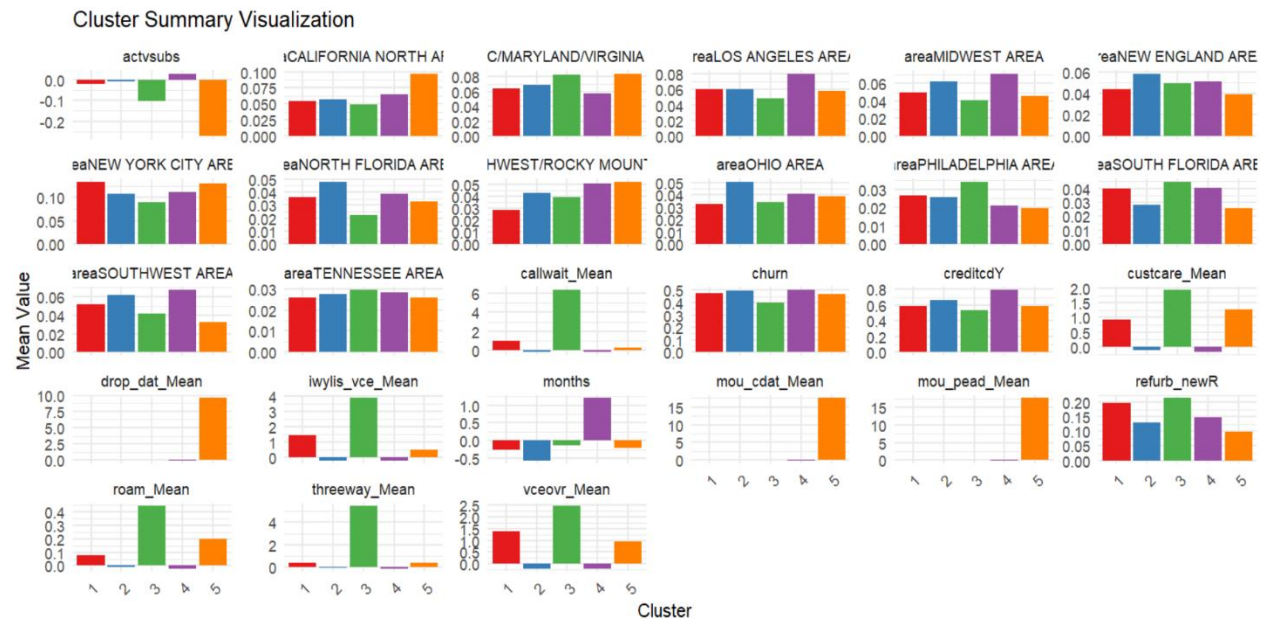
4. Clustering Analysis:

Usage: Clustering is a technique that involves organizing a collection of items in a manner where items within the same group exhibit greater similarity to each other compared to items in different groups.

Application in Churn Analysis: Used to segment the customer base into distinct groups based on their characteristics and behaviors.

Strengths: Helps in identifying natural groupings within the data, useful for targeted marketing and personalized customer retention strategies.

Limitations: Determining the number of clusters can be subjective, and the outcome depends on the chosen method and parameters.



```
## # A tibble: 5 × 28
##   cluster churn actvsubs callwait_Mean custcare_Mean iwylis_vce_Mean months
##   <fct>   <dbl>   <dbl>         <dbl>         <dbl>         <dbl> <dbl>
## 1 1      0.475 -0.0219      0.963         0.920         1.42  -0.268
## 2 2      0.497 -0.00823    -0.185        -0.111        -0.216 -0.562
## 3 3      0.398 -0.102      6.34          1.94          3.88  -0.145
## 4 4      0.503  0.0285     -0.177        -0.187        -0.217  1.20
## 5 5      0.468 -0.270      0.260         1.28          0.470 -0.221
## # i 21 more variables: mou_cdat_Mean <dbl>, mou_pead_Mean <dbl>,
## #   roam_Mean <dbl>, threeway_Mean <dbl>, vceovr_Mean <dbl>,
## #   `areaCALIFORNIA NORTH AREA` <dbl>, `areaDC/MARYLAND/VIRGINIA AREA` <dbl>,
## #   `areaLOS ANGELES AREA` <dbl>, `areaMIDWEST AREA` <dbl>,
## #   `areaNEW ENGLAND AREA` <dbl>, `areaNEW YORK CITY AREA` <dbl>,
## #   `areaNORTH FLORIDA AREA` <dbl>, `areaNORTHWEST/ROCKY MOUNTAIN AREA` <dbl>,
## #   `areaOHIO AREA` <dbl>, `areaPHILADELPHIA AREA` <dbl>, ...
```

Cluster Overview and Strategies to be implemented:

1. Cluster 1 High Engagement, Moderate Churn:

Characteristics: Moderate churn rate, high engagement in services like call waiting and customer care.

Strategy: Target with loyalty programs and personalized offers. Upsell highertier services or bundles, emphasizing customer service quality.

2. Cluster 2 Average Engagement, High Churn:

Characteristics: Near average subscriptions, lower engagement in additional services, high churn rate.

Strategy: Focus on retention strategies. Improve engagement through targeted promotions on underused services. Conduct surveys to understand the cause of dissatisfaction.

3. Cluster 3 Selective Engagement, Lower Churn:

Characteristics: Very high engagement in certain services, notably lower churn rate.

Strategy: Ideal candidates for premium services. Engage with exclusive offers and premium support options. Explore cross selling opportunities.

4. Cluster 4 Low Engagement, Highest Churn:

Characteristics: Slightly higher subscriptions but lower engagement in services, highest churn rate.

Strategy: Critical segment for retention efforts. Implement win back campaigns and offer special discounts or trial periods for additional services. Analyze service usage patterns for more tailored offerings.

5. Cluster 5 Diverse Engagement, Moderate Churn:

Characteristics: Lowest subscriptions, diverse service usage, moderate churn.

Marketing Strategy: Focus on increasing subscriptions through trial offers and bundling. Enhance engagement through targeted communication highlighting service benefits.

Conclusion:

Leveraging cluster analysis for marketing segmentation allows for a more nuanced understanding of the customer base. By recognizing and responding to the distinct needs and behaviors of each cluster,

businesses can deploy more effective marketing strategies, enhance customer satisfaction, and ultimately drive growth and profitability.

Recommendations:

- 1. Implement targeted marketing campaigns for each cluster.
- 2. Continuously monitor and reassess cluster dynamics and adjust strategies accordingly.
- 3. Invest in data analytics capabilities to refine clustering methods and uncover deeper insights.
- 4. Engage with customers within each cluster to validate strategies and gather feedback for ongoing improvement.

References

Appendix:

Data Dictionary:

The document provides detailed descriptions of various variables in the telecom customer churn dataset. Each variable plays a specific role in understanding customer behavior and predicting churn in the telecom industry.

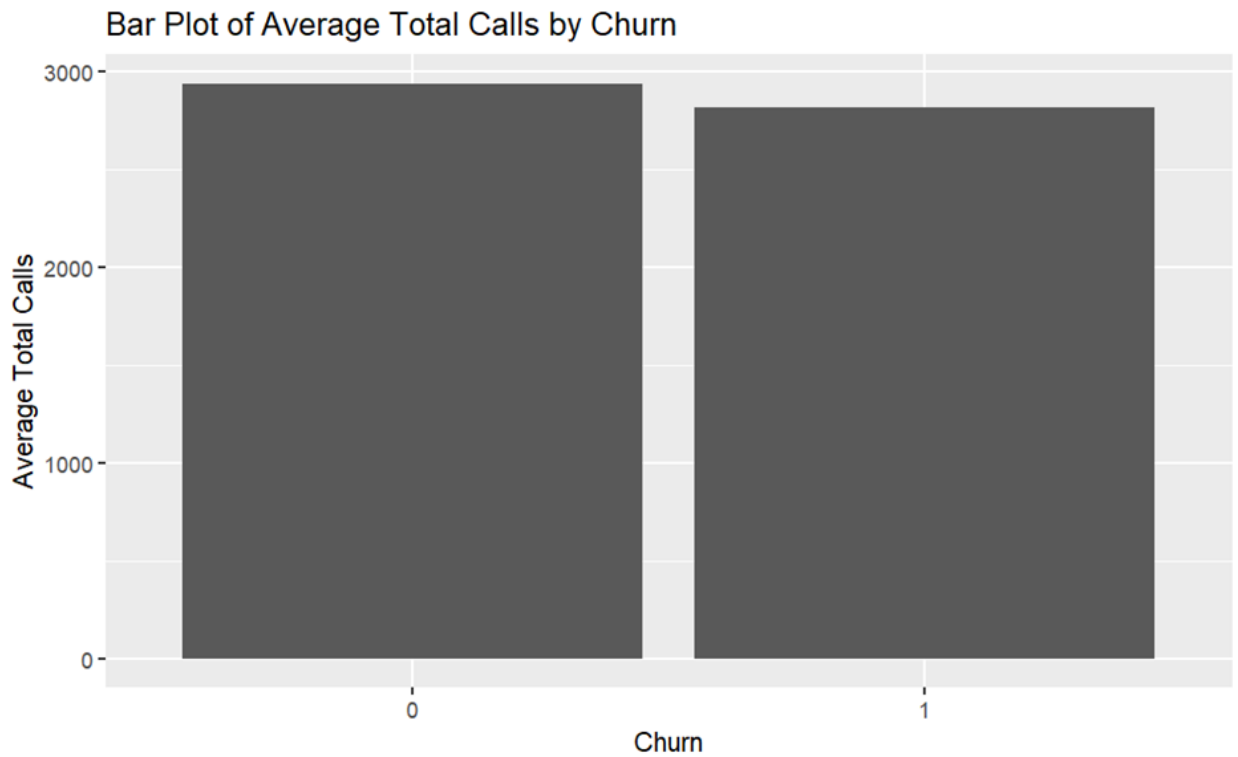
Variable Name	Description
actvsubs	Number of active subscribers in household
area	Geographic area
attempt_Mean	Mean number of attempted calls
blk_dat_Mean	Mean number of blocked (failed) data calls
blk_vce_Mean	Mean number of blocked (failed) voice calls
callfwdv_Mean	Mean number of call forwarding calls
callwait_Mean	Mean number of call waiting calls
churn	Instance of churn between 3160 days after observation date
comp_dat_Mean	Mean number of completed data calls
comp_vce_Mean	Mean number of completed voice calls

complete_Mean	Mean number of completed calls
creditcd	Credit card indicator
custcare_Mean	Mean number of customer care calls
da_Mean	Mean number of directory assisted calls
datovr_Mean	Mean revenue of data overage
drop_blk_Mean	Mean number of dropped or blocked calls
drop_dat_Mean	Mean number of dropped (failed) data calls
drop_vce_Mean	Mean number of dropped (failed) voice calls
inonemin_Mean	Mean number of inbound calls less than one minute
iwylis_vce_Mean	Mean number of inbound wireless to wireless voice calls
months	Total number of months in service
mou_cdat_Mean	Mean unrounded minutes of use of completed data calls
mou_cvce_Mean	Mean unrounded minutes of use of completed voice calls
mou_opkd_Mean	Mean unrounded minutes of use of offpeak data calls

Report on Selection Criteria for Telecom Dataset Variable

Literature Review Links

1. Hindawi. (n.d.). Detecting the risk of customer churn in telecom sector: A comparative study. Mathematical Problems in Engineering. Retrieved from [\[URL\]](#)
- 2.. <https://bmcbioinformatics.biomedcentral.com/articles/10.1186/s128590182448z>
3. McKinsey & Company. (n.d.). Reducing churn in telecom through advanced analytics. Retrieved from [\[URL\]](#)



```
# List of skewed numerical variables
skewed_vars <- c("da_Mean", "datovr_Mean", "ovrmou_Mean", "ovrrev_Mean", "roam_Mean", "vceovr_Mean")

# Impute missing values with the median for each variable
for (var in skewed_vars) {
  # Calculate the median excluding NA values
  median_value <- median(churn_dataset[[var]], na.rm = TRUE)
  # Replace NA with the median value
  churn_dataset[[var]][is.na(churn_dataset[[var]])] <- median_value
}
```

```
print(missing_values)
```

```
##      actvsubs      area  attempt_Mean  blk_dat_Mean  blk_vce_Mean
##      0          40          0          0          0
##  callfwdv_Mean  callwait_Mean      churn  comp_dat_Mean  comp_vce_Mean
##      0          0          0          0          0
##  complete_Mean  creditcd  custcare_Mean      da_Mean  datovr_Mean
##      0        1732          0        357        357
##  drop_blk_Mean  drop_dat_Mean  drop_vce_Mean  inonemin_Mean  iwylis_vce_Mean
##      0          0          0          0          0
##      months  mou_cdat_Mean  mou_cvce_Mean  mou_opkd_Mean  mou_opkv_Mean
##      0          0          0          0          0
##  mou_pead_Mean  mou_peav_Mean  mou_rvce_Mean  opk_dat_Mean  opk_vce_Mean
##      0          0          0          0          0
##  ovrrou_Mean  ovrrev_Mean  owylis_vce_Mean  peak_dat_Mean  peak_vce_Mean
##      357        357          0          0          0
##  plcd_dat_Mean  plcd_vce_Mean  recv_sms_Mean  recv_vce_Mean  refurb_new
##      0          0          0          0          1
##      roam_Mean  threeway_Mean      totcalls      totmou      totrev
##      357          0          0          0          0
##  unan_dat_Mean  unan_vce_Mean  vceovr_Mean
##      0          0        357
```

Handling of Missing Values and creation of dummies for Categorical variables

```
# Function to calculate mode
get_mode <- function(v) {
  univq <- unique(v)
  univq[which.max(tabulate(match(v, univq)))]
}

# Step 1: Handle missing data
churn_dataset$area[is.na(churn_dataset$area)] <- get_mode(churn_dataset$area)
churn_dataset$creditcd[is.na(churn_dataset$creditcd)] <- get_mode(churn_dataset$creditcd)
churn_dataset$refurb_new[is.na(churn_dataset$refurb_new)] <- get_mode(churn_dataset$refurb_new)

# Step 2: Create dummy variables
# Note: This process automatically drops one level for each factor
churn_dataset <- cbind(churn_dataset, model.matrix(~ area + creditcd + refurb_new - 1, data = churn_dataset))

# Optionally, you might want to remove the original columns to avoid redundancy
churn_dataset <- churn_dataset[, !colnames(churn_dataset) %in% c("area", "creditcd", "refurb_new")]

# View the updated dataset
head(churn_dataset)
```

Deleted Variables:

Sl.no	Name of the Variable
1	attempt Mean
2	blk vce Mean
3	comp vce Mean
4	inonemin Mean

5	mou cvce Mean
6	mou opkv Mean
7	mou peav Mean
8	opk vce Mean
9	peak vce Mean
10	plcd dat Mean
11	Totcalls
12	Totmou
13	Totrev
14	unan vce Mean